

fx_shape3_fl

\$FL Phase 39.FL D3 post-flash cycle FAIL forensic + root-cause fixes — Completed (→ Fixed.md) (User mandate 2026-05-20) [migrated from Issues.md 2026-05-27]

Status: Completed (→ Fixed.md) **Type:** Task **Forensic anchor:**
D3 cycle qa-results/d3_cycle_20260520T0507Z/test_all_fixes.log re-
corded 14 unique FAIL signatures (29 phase-rep FAILs). User
mandate: "Do no. 2 but make sure that all failures are properly in-
vestigated and fully fixed with root causes being treated prop-
erly!"

**Per-FAIL classification + action (per §11.4.42 systematic-de-
bugging):**

#	FAIL	Classification	Root cause	A
1	NanoK- VM In- tegration	TEST-INFRASTRUCTURE-BUG	set -e + x=\$(creds_get ...) non-zero exit kills script before [G]/[H]/[I] sections run when / data/local/tmp/ secrets/.nanokvm.env absent	A in t
2	\$BJ Fix #135 SW H.264 Encoder Fallback	TEST-INFRASTRUCTURE-BUG (2 issues)	(a) ab_section missing from anti_bluff.sh; (b) ab_skip required \$2 un- der set -u; (c) B1 pars- er used od -c (space- separated output) + grep "ftyp mp42" (no- space pattern) — mis- match made EVERY mp4 fail validation	A m b
3-6	§11.4.58 PWU val- idation	TEST-INFRASTRUCTURE-BUG (single root)	_ab_strip_volatile did not normalise ACTION #N counter values or com-	E A t

#	FAIL	Classification	Root cause	A
	surfaces (4 tests)		pact 12-char hex tokens — every iteration produced different hash → divergence FAIL per §11.4.50	
7	Hardware Video Codecs	KNOWN-DISABLED-BY-DESIGN	Fix #135 \$BJ disabled c2.rk.avc.encoder per Phase 39.EN; test was missing the §107-equivalent encoder-disable acceptance branch	U S d #
8-11	§O per-app video FAILs (YouTube, Kodi, VLC, AIMP)	TEST-INFRASTRUCTURE-BUG (single root)	per_app_video_harness.sh fallback only fired on err=-38 stderr signature. After Fix #135 disabled HW encoder, SW path produces small clips with EMPTY stderr — fallback never fired → wc -l < frame_manifest.tsv failed → phantom FAIL	V A e
12	D9 orientation steady-state (Pavel #4)	TEST-INFRASTRUCTURE-BUG (stale expectation)	Phase 39.BV (§AW fix) relaxed watchdog INTERVAL_SEC 1→60 (safety-net mode; primary correction migrated to PresenterService RotationLockObserver sub-50ms). Test still expected old INTERVAL_SEC=1	A 3
13	Lampa + TorrServe E2E	PENDING_FORENSICS:operator-attended	TorrServe service alive but :8090 LISTEN socket never bound (UI tap on "Start server" required by upstream — NOT a Fix #120 regression). dumphsys activity services shows TorrServe running, /proc/<pid>/net/tcp shows zero LISTEN	D w T n S
14	Streaming-app stress	PENDING_FORENSICS (intermittent)	S3b OCR-classify unknown branch fails-open per §11.4.6 (correct conservative choice). Phase 3 PASSEd (0 FAIL), Phase	T fi c §

#	FAIL	Classification	Root cause	A
			1+2 saw 2-3 FAILs — captured-evidence-confirmed flakiness in RuTube transient loading-state frames. NOT a regression of OUR fix	

Files modified:

- device/rockchip/rk3588/tests/test_nanokvm_integration.sh
- device/rockchip/rk3588/tests/lib/anti_bluff.sh (ab_skip tolerance + ab_section helper + strip-volatile patterns)
- device/rockchip/rk3588/tests/test_bj_sw_encode_fallback.sh
- device/rockchip/rk3588/tests/test_hw_video_codecs.sh
- device/rockchip/rk3588/tests/per_app_video/per_app_video_harness.sh
- device/rockchip/rk3588/tests/test_orientation_d9_steady_state.sh
- device/rockchip/rk3588/tests/test_lampa_torrserve_e2e_secondary.sh

Live-ADB validation: All 13 actionable FAILs verified PASS on D3 (998fd36615e99484) via direct test execution per §11.4.51 LIVE_ADB_TESTABLE classification (test-script-only fixes; no rebuild required).

Captured-evidence: qa-results/d3_cycle_20260520T0507Z/test_all_fixes.log (pre-fix), per-test on-device re-run logs (post-fix, in this session's tool-call transcript).

§11.4.6 no-guessing classification: FAIL #14 explicitly marked PENDING_FORENSICS: — captured intermittent across cycle phases; root cause UNCONFIRMED beyond "RuTube transient state". No demotion-evidence claim per §11.4.7.